SQL PROJECT

Current Table - 1067371 rows

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer_ID	Country
1	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	2009-12-01 00:00:00.0000000	6.95	13085	United Kingdom
2	489434	79323P	PINK CHERRY LIGHTS	12	2009-12-01 00:00:00.0000000	6.75	13085	United Kingdom
3	489434	79323W	WHITE CHERRY LIGHTS	12	2009-12-01 00:00:00.0000000	6.75	13085	United Kingdom
4	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	2009-12-01 00:00:00.0000000	2.1	13085	United Kingdom
5	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	2009-12-01 00:00:00.0000000	1.25	13085	United Kingdom
6	489434	22064	PINK DOUGHNUT TRINKET POT	24	2009-12-01 00:00:00.0000000	1.65	13085	United Kingdom
7	489434	21871	SAVE THE PLANET MUG	24	2009-12-01 00:00:00.0000000	1.25	13085	United Kingdom
8	489434	21523	FANCY FONT HOME SWEET HOME DOORMAT	10	2009-12-01 00:00:00.0000000	5.95	13085	United Kingdom
9	489435	22350	CAT BOWL	12	2009-12-01 00:00:00.0000000	2.55	13085	United Kingdom
10	489435	22349	DOG BOWL , CHASING BALL DESIGN	12	2009-12-01 00:00:00.0000000	3.75	13085	United Kingdom
11	489435	22195	HEART MEASURING SPOONS LARGE	24	2009-12-01 00:00:00.0000000	1.65	13085	United Kingdom
12	489435	22353	LUNCHBOX WITH CUTLERY FAIRY CAKES	12	2009-12-01 00:00:00.0000000	2.55	13085	United Kingdom
13	489436	48173C	DOOR MAT BLACK FLOCK	10	2009-12-01 00:00:00.0000000	5.95	13078	United Kingdom
14	489436	21755	LOVE BUILDING BLOCK WORD	18	2009-12-01 00:00:00.0000000	5.45	13078	United Kingdom
15	489436	21754	HOME BUILDING BLOCK WORD	3	2009-12-01 00:00:00.0000000	5.95	13078	United Kingdom
16	489436	84879	ASSORTED COLOUR BIRD ORNAMENT	16	2009-12-01 00:00:00.0000000	1.69	13078	United Kingdom
17	489436	22119	PEACE WOODEN BLOCK LETTERS	3	2009-12-01 00:00:00.0000000	6.95	13078	United Kingdom
18	489436	22142	CHRISTMAS CRAFT WHITE FAIRY	12	2009-12-01 00:00:00.0000000	1.45	13078	United Kingdom
19	489436	22296	HEART IVORY TRELLIS LARGE	12	2009-12-01 00:00:00.0000000	1.65	13078	United Kingdom
20	489436	22295	HEART FILIGREE DOVE LARGE	12	2009-12-01 00:00:00.0000000	1.65	13078	United Kingdom
21	489436	22109	FULL ENGLISH BREAKFAST PLATE	16	2009-12-01 00:00:00.0000000	3.39	13078	United Kingdom
22	489436	22107	PIZZA PLATE IN BOX	4	2009-12-01 00:00:00.0000000	3.75	13078	United Kingdom
23	489436	22194	BLACK DINER WALL CLOCK	2	2009-12-01 00:00:00.0000000	8.5	13078	United Kingdom
24	489436	35004B	SET OF 3 BLACK FLYING DUCKS	12	2009-12-01 00:00:00.0000000	4.65	13078	United Kingdom
25	489436	82582	AREA PATROLLED METAL SIGN	12	2009-12-01 00:00:00.0000000	2.1	13078	United Kingdom
26	489436	21181	PLEASE ONE PERSON METAL SIGN	12	2009-12-01 00:00:00.0000000	2.1	13078	United Kingdom
27	489436	21756	BATH BUILDING BLOCK WORD	3	2009-12-01 00:00:00.0000000	5.95	13078	United Kingdom
28	489436	21333	CLASSIC WHITE FRAME	6	2009-12-01 00:00:00.0000000	2.95	13078	United Kingdom
29	489436	84596F	SMALL MARSHMALLOWS PINK BOWL	8	2009-12-01 00:00:00.0000000	1.25	13078	United Kingdom
30	489436	84596L	BISCUITS SMALL BOWL LIGHT BLUE	8	2009-12-01 00:00:00.0000000	1.25	13078	United Kingdom
31	489436	22111	SCOTTIE DOG HOT WATER BOTTLE	24	2009-12-01 00:00:00.0000000	4.25	13078	United Kingdom

PHASE1Checking Null Values

```sql
SELECT COUNT(*)-COUNT(Description)
FROM  [dbo].[online_retail_II]
--4382 nulls

SELECT COUNT(*)-COUNT(Quantity)
FROM  [dbo].[online_retail_II]
--0

SELECT COUNT(*)-COUNT(InvoiceDate)
FROM  [dbo].[online_retail_II]
--0

SELECT COUNT(*)-COUNT(Price)
FROM  [dbo].[online_retail_II]
--0
SELECT COUNT(*)-COUNT(Customer_ID)
FROM  [dbo].[online_retail_II]
--243007

SELECT COUNT(*)-COUNT(Country)
FROM  [dbo].[online_retail_II]
--0
```

Populating Description- Nulls

```sql
--step1-check
SELECT  a.StockCode, ISNULL(a.Description, b.Description)AS NULL_DESCRIPTION,
       a.Description AS MissingDescription,
       b.Description AS AvailableDescription
FROM dbo.online_retail_II AS a
JOIN dbo.online_retail_II AS b
ON a.StockCode = b.StockCode
WHERE a.Description IS NULL
AND b.Description IS NOT NULL;

--step-2alter
--NOT creating a separate table

--step3-update
UPDATE A
SET Description= ISNULL(a.Description, b.Description)
FROM dbo.online_retail_II AS a
JOIN dbo.online_retail_II AS b
ON a.StockCode = b.StockCode
WHERE a.Description IS NULL
AND b.Description IS NOT NULL;
```

Removing Duplicates

```sql
WITH CTE_DUPLI AS
(
SELECT [Invoice],
ROW_NUMBER() OVER(PARTITION BY [Invoice],
[StockCode],[Description],[Quantity], [InvoiceDate],
[Price], [Customer_ID],[Country] ORDER BY INVOICE  ) AS DUPLI
FROM [dbo].[online_retail_II]
)

SELECT INVOICE, DUPLI
FROM CTE_DUPLI
WHERE DUPLI >1


--step-2alter

WITH CTE_DUPLI AS
(
SELECT [Invoice],
ROW_NUMBER() OVER(PARTITION BY [Invoice],
[StockCode],[Description],[Quantity], [InvoiceDate],
[Price], [Customer_ID],[Country] ORDER BY INVOICE  ) AS DUPLI
FROM [dbo].[online_retail_II]
)

DELETE
FROM CTE_DUPLI
WHERE DUPLI>1
```

Change Date Format

```sql
--step1-check
SELECT INVOICE, INVOICEDATE, CONVERT(DATE, INVOICEDATE) AS CONVERT_DATE
FROM [dbo].[online_retail_II]

--ALTER

ALTER TABLE [dbo].[online_retail_II]
ADD N_DATE DATE;

--UPDATE
UPDATE [dbo].[online_retail_II]
SET N_DATE= CONVERT(DATE, INVOICEDATE)

--ALTER

ALTER TABLE [dbo].[online_retail_II]
DROP COLUMN [InvoiceDate]
```

	Invoice	StockCode	Description	Quantity	Price	Customer_ID	Country	N_DATE
1	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	6.95	13085	United Kingdom	2009-12-01
2	489434	79323P	PINK CHERRY LIGHTS	12	6.75	13085	United Kingdom	2009-12-01
3	489434	79323W	WHITE CHERRY LIGHTS	12	6.75	13085	United Kingdom	2009-12-01
4	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	2.1	13085	United Kingdom	2009-12-01
5	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	1.25	13085	United Kingdom	2009-12-01
6	489434	22064	PINK DOUGHNUT TRINKET POT	24	1.65	13085	United Kingdom	2009-12-01
7	489434	21871	SAVE THE PLANET MUG	24	1.25	13085	United Kingdom	2009-12-01
8	489434	21523	FANCY FONT HOME SWEET HOME DOORMAT	10	5.95	13085	United Kingdom	2009-12-01
9	489435	22350	CAT BOWL	12	2.55	13085	United Kingdom	2009-12-01
10	489435	22349	DOG BOWL , CHASING BALL DESIGN	12	3.75	13085	United Kingdom	2009-12-01
11	489435	22195	HEART MEASURING SPOONS LARGE	24	1.65	13085	United Kingdom	2009-12-01
12	489435	22353	LUNCHBOX WITH CUTLERY FAIRY CAKES	12	2.55	13085	United Kingdom	2009-12-01
13	489436	48173C	DOOR MAT BLACK FLOCK	10	5.95	13078	United Kingdom	2009-12-01

Create Calculated Fields

```sql
ALTER TABLE [dbo].[online_retail_II]
ADD TOTAL_SALES DECIMAL(10,2)

UPDATE  [dbo].[online_retail_II]
SET TOTAL_SALES= QUANTITY* PRICE

SELECT * FROM [dbo].[online_retail_II]
```

	Invoice	StockCode	Description	Quantity	Price	Customer_ID	Country	N_DATE	TOTAL_SALES
1	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	6.95	13085	United Kingdom	2009-12-01	83.40
2	489434	79323P	PINK CHERRY LIGHTS	12	6.75	13085	United Kingdom	2009-12-01	81.00
3	489434	79323W	WHITE CHERRY LIGHTS	12	6.75	13085	United Kingdom	2009-12-01	81.00
4	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	2.1	13085	United Kingdom	2009-12-01	100.80
5	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	1.25	13085	United Kingdom	2009-12-01	30.00
6	489434	22064	PINK DOUGHNUT TRINKET POT	24	1.65	13085	United Kingdom	2009-12-01	39.60
7	489434	21871	SAVE THE PLANET MUG	24	1.25	13085	United Kingdom	2009-12-01	30.00
8	489434	21523	FANCY FONT HOME SWEET HOME DOORMAT	10	5.95	13085	United Kingdom	2009-12-01	59.50
9	489435	22350	CAT BOWL	12	2.55	13085	United Kingdom	2009-12-01	30.60
10	489435	22349	DOG BOWL , CHASING BALL DESIGN	12	3.75	13085	United Kingdom	2009-12-01	45.00
11	489435	22195	HEART MEASURING SPOONS LARGE	24	1.65	13085	United Kingdom	2009-12-01	39.60
12	489435	22353	LUNCHBOX WITH CUTLERY FAIRY CAKES	12	2.55	13085	United Kingdom	2009-12-01	30.60
13	489436	48173C	DOOR MAT BLACK FLOCK	10	5.95	13078	United Kingdom	2009-12-01	59.50
14	489436	21755	LOVE BUILDING BLOCK WORD	18	5.45	13078	United Kingdom	2009-12-01	98.10
15	489436	21754	HOME BUILDING BLOCK WORD	3	5.95	13078	United Kingdom	2009-12-01	17.85
16	489436	84879	ASSORTED COLOUR BIRD ORNAMENT	16	1.69	13078	United Kingdom	2009-12-01	27.04
17	489436	22119	PEACE WOODEN BLOCK LETTERS	3	6.95	13078	United Kingdom	2009-12-01	20.85
18	489436	22142	CHRISTMAS CRAFT WHITE FAIRY	12	1.45	13078	United Kingdom	2009-12-01	17.40
19	489436	22296	HEART IVORY TRELLIS LARGE	12	1.65	13078	United Kingdom	2009-12-01	19.80

PHASE2: Regional Sales

```
SELECT COUNTRY, SUM(TOTAL_SALES) AS REGION_SALES
FROM [dbo].[online_retail_II]
WHERE QUANTITY>0
GROUP BY COUNTRY
ORDER BY REGION_SALES DESC
```

	COUNTRY	REGION_SALES
1	United Kingdom	17251859.93
2	EIRE	658767.31
3	Netherlands	554038.09
4	Germany	425019.71
5	France	350456.09
6	Australia	169283.46
7	Spain	108332.49
8	Switzerland	100685.59
9	Sweden	91869.82
10	Denmark	68580.69
11	Belgium	65387.82
12	Norway	56322.50
13	Portugal	56283.53
14	Channel Islands	44623.33
15	Japan	43023.91
16	Italy	32108.17
17	Finland	29925.54
18	Singapore	25317.06
19	Cyprus	24849.95
20	Hong Kong	23685.52

Time Based Trends- Per month

```
SELECT MONTH_T, MONTH(N_DATE) month_num ,  SUM(sum(TOTAL_SALES)) OVER(ORDER BY MONTH_T)AS month_sum
FROM (SELECT N_DATE,TOTAL_SALES, DATENAME(MONTH,N_DATE)AS MONTH_T
FROM  [dbo].[online_retail_II]
 )T
 GROUP BY MONTH_T, MONTH(N_DATE)
 ORDER BY MONTH(N_DATE)
```

	MONTH_T	month_num	month_sum
1	January	1	6601736.68
2	February	2	5420808.62
3	March	3	10667539.68
4	April	4	1080294.53
5	May	5	12002904.50
6	June	6	9222278.46
7	July	7	7855227.36
8	August	8	2416455.38
9	September	9	18855500.08
10	October	10	16986801.34
11	November	11	14875747.50
12	December	12	4392516.85

Product Performance

```sql
WITH CTE_PRODCUTPERFORMANCE AS
(
SELECT STOCKCODE, sum(total_sales) sales_total, count(*) as stock_count
FROM (SELECT STOCKCODE, TOTAL_SALES, QUANTITY
FROM [dbo].[online_retail_II]) T
WHERE QUANTITY>0
group by STOCKCODE
)
SELECT *, dense_rank() over(order by sales_total desc) as performance_ranks
FROM CTE_PRODCUTPERFORMANCE
WHERE SALES_TOTAL> (SELECT AVG(SALES_TOTAL) FROM CTE_PRODCUTPERFORMANCE)
```

	STOCKCODE	sales_total	stock_count	performance_ranks
1	M	339614.86	859	1
2	22423	330590.32	3954	2
3	DOT	309854.11	1422	3
4	85123A	261168.73	5616	4
5	85099B	182680.98	4072	5
6	23843	168469.60	1	6
7	47566	148318.28	2708	7
8	84879	129324.49	2849	8
9	POST	125682.42	1858	9
10	22086	117760.29	2069	10
11	23166	81700.92	250	11
12	79321	80540.88	1156	12
13	22197	79520.20	2373	13
14	22386	76324.62	2269	14
15	84347	71300.40	784	15
16	20725	71106.29	3090	16
17	85099F	69002.13	1879	17
18	48138	67010.31	1670	18
19	23084	66870.03	1030	19
20	20685	66417.47	1732	20
21	21137	65189.29	636	21
22	21843	64918.30	1530	22
23	21931	61537.55	2353	23
24	22502	60880.30	941	24
25	21621	60286.04	1308	25

Customer Behaviour Analysis

```sql
WITH CUSTOMER_BEHAVIOUR AS
(
SELECT CUSTOMER_ID, COUNT(*) NUM_PURCHASE, SUM(TOTAL_SALES) AS SALES_CUSTOMER
FROM[dbo].[online_retail_II]
WHERE QUANTITY>0 AND CUSTOMER_ID IS NOT NULL
GROUP BY CUSTOMER_ID
)

,PURCHASE_RANK AS
(
SELECT CUSTOMER_ID, DENSE_RANK() OVER(ORDER BY NUM_PURCHASE DESC) AS PURCHASE_RNK
FROM CUSTOMER_BEHAVIOUR
)


SELECT a.* ,b.PURCHASE_RNK, (CASE
WHEN PURCHASE_RNK BETWEEN 1 AND 228 THEN 'HIGH PURCHASE CUSTOMER'
WHEN PURCHASE_RNK BETWEEN 229 AND  THEN 'MEDIUM PURCHASE CUSTOMER'
ELSE 'LOW PURCHASE CUSTOMER'
END) AS CUSTOMER_BEHAVIOUR
FROM CUSTOMER_BEHAVIOUR AS A
JOIN PURCHASE_RANK AS B
ON A.CUSTOMER_ID= B.CUSTOMER_ID
order by  PURCHASE_RNK
```

	CUSTOMER_ID	NUM_PURCHASE	SALES_CUSTOMER	PURCHASE_RNK	CUSTOMER_BEHAVIOUR
1	17841	12435	68545.25	1	HIGH PURCHASE CUSTOMER
2	14911	11079	291420.81	2	HIGH PURCHASE CUSTOMER
3	12748	6587	53539.64	3	HIGH PURCHASE CUSTOMER
4	14606	6359	29951.37	4	HIGH PURCHASE CUSTOMER
5	14096	5111	65164.79	5	HIGH PURCHASE CUSTOMER
6	15311	4306	114966.42	6	HIGH PURCHASE CUSTOMER
7	14156	4038	313437.62	7	HIGH PURCHASE CUSTOMER
8	14646	3854	528602.52	8	HIGH PURCHASE CUSTOMER
9	13089	3316	113416.91	9	HIGH PURCHASE CUSTOMER
10	16549	3093	12781.80	10	HIGH PURCHASE CUSTOMER
11	14298	2864	91194.49	11	HIGH PURCHASE CUSTOMER
12	15039	2695	39141.48	12	HIGH PURCHASE CUSTOMER
13	14527	2639	27246.88	13	HIGH PURCHASE CUSTOMER
14	17850	2499	51208.87	14	HIGH PURCHASE CUSTOMER
15	15005	2387	13754.58	15	HIGH PURCHASE CUSTOMER
16	13081	2284	56838.37	16	HIGH PURCHASE CUSTOMER
17	13263	1905	8676.20	17	HIGH PURCHASE CUSTOMER
18	17511	1868	172132.87	18	HIGH PURCHASE CUSTOMER
19	14159	1836	7860.80	19	HIGH PURCHASE CUSTOMER
20	15719	1770	10195.97	20	HIGH PURCHASE CUSTOMER
21	17377	1712	17644.27	21	HIGH PURCHASE CUSTOMER
22	16782	1683	9636.12	22	HIGH PURCHASE CUSTOMER
23	17920	1640	23581.78	23	HIGH PURCHASE CUSTOMER

Month On Month Sales

--((Current Period Value - Previous Period Value) / Previous Period Value) * 100

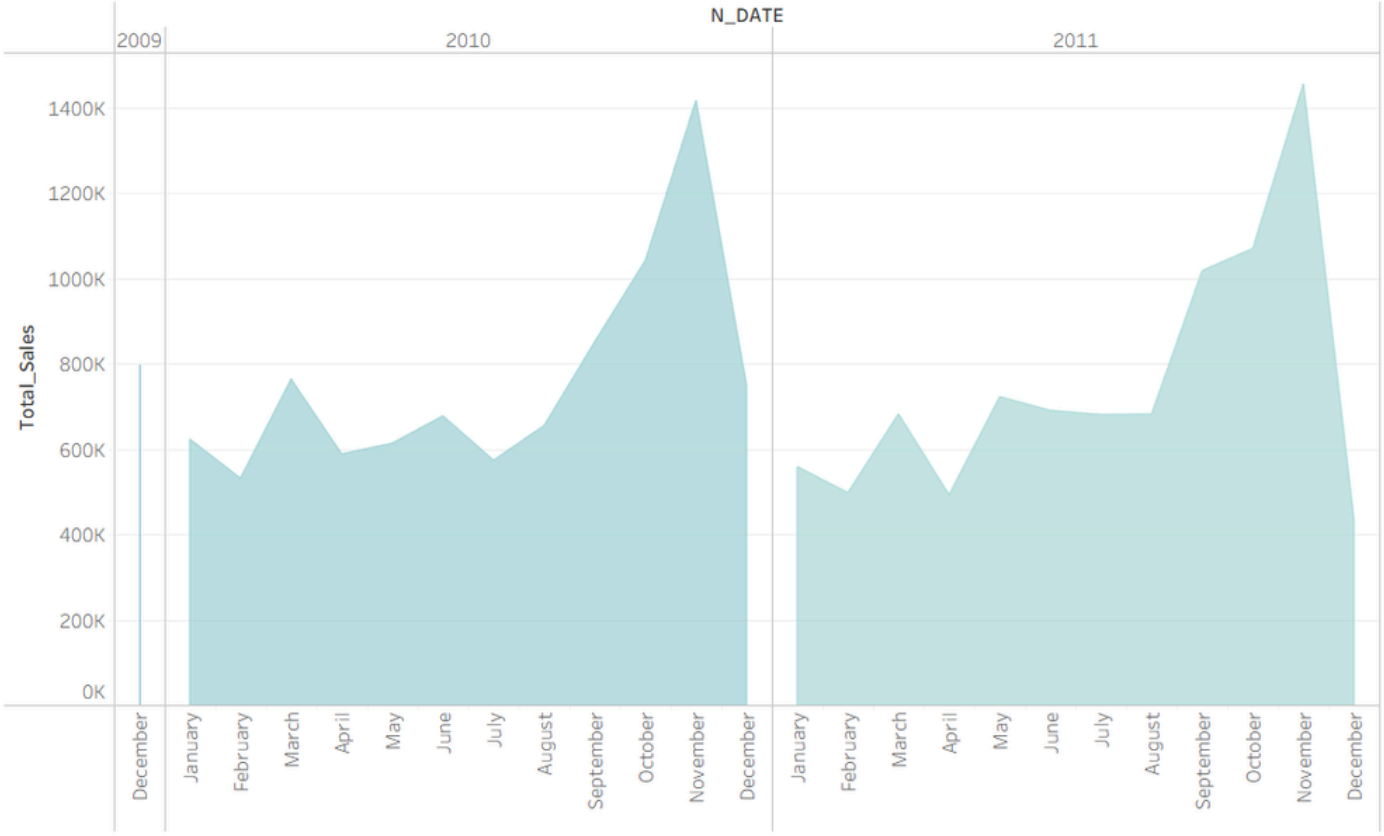
```sql
WITH MONTH_CTE AS
(
SELECT LAG(SUM_SALES) OVER(ORDER BY N_YEAR,  N_MONTH ) AS SUM_PRV, N_MONTH, N_YEAR, SUM_SALES
FROM (SELECT MONTH(N_DATE) AS N_MONTH, YEAR(N_DATE) AS N_YEAR, SUM(TOTAL_SALES) SUM_SALES
FROM [dbo].[online_retail_II]
WHERE Customer_ID IS NOT NULL AND QUANTITY >0
GROUP BY MONTH(N_DATE), YEAR(N_DATE) )T
)


SELECT N_MONTH,N_YEAR,SUM_SALES, ( ((SUM_SALES- SUM_PRV)/NULLIF(SUM_PRV,0))*100 )AS MONTH_ON_MONTH_SALES
FROM MONTH_CTE
WHERE SUM_PRV>0
```

	N_MONTH	N_YEAR	SUM_SALES	MONTH_ON_MONTH_SALES
1	1	2010	555802.67	-18.679800
2	2	2010	504558.95	-9.219700
3	3	2010	696978.47	38.136100
4	4	2010	591982.00	-15.064500
5	5	2010	597833.38	0.988400
6	6	2010	636371.13	6.446200
7	7	2010	589736.17	-7.328200
8	8	2010	602224.60	2.117600
9	9	2010	829013.95	37.658500
10	10	2010	1033112.01	24.619300
11	11	2010	1166460.02	12.907400
12	12	2010	570422.73	-51.097900
13	1	2011	568101.31	-0.406900
14	2	2011	446084.92	-21.477900
15	3	2011	594081.76	33.176800
16	4	2011	468374.33	-21.159900
17	5	2011	677355.15	44.618300
18	6	2011	660046.05	-2.555300
19	7	2011	598962.90	-9.254300
20	8	2011	644051.04	7.527700
21	9	2011	950686.30	47.610300
22	10	2011	1035642.45	8.936200

Phase 3: Key Sales Trends and Patterns

Key sales trends and patterns.



In the years 2010 and 2011 evident patterns based on the sales can be observed. The sales peaked in November and then faced a sudden drop by the end of December. The year end is not the period of great advantage, although November is the time the stock has to be maxed. August to October is the mid-sales period, were the tread grows steadily.

Top Performing products and regions

The top 5 items with the highest sales are the following

DOTCOM POSTAGE

REGENCY CAKESTAND 3 TIER

Manual

WHITE HANGING HEART T-LIGHT HOLDER

PAPER CRAFT , LITTLE BIRDIE

```sql
WITH CTE_PRODCUTPERFORMANCE AS
(
SELECT STOCKCODE, sum(total_sales) sales_total, count(*) as stock_count, COUNTRY, DESCRIPTION
FROM (SELECT STOCKCODE, TOTAL_SALES, QUANTITY, COUNTRY, DESCRIPTION
FROM [dbo].[online_retail_II]) T
WHERE QUANTITY>0
group by STOCKCODE, COUNTRY, DESCRIPTION
)
SELECT *, dense_rank() over(order by sales_total desc) as performance_ranks
FROM CTE_PRODCUTPERFORMANCE
WHERE SALES_TOTAL> (SELECT AVG(SALES_TOTAL) FROM CTE_PRODCUTPERFORMANCE)
```

	STOCKCODE	sales_total	stock_count	COUNTRY	DESCRIPTION	performance_ranks
1	DOT	309854.11	1422	United Kingdom	DOTCOM POSTAGE	1
2	22423	277783.12	3472	United Kingdom	REGENCY CAKESTAND 3 TIER	2
3	M	244853.06	758	United Kingdom	Manual	3
4	85123A	240796.72	5399	United Kingdom	WHITE HANGING HEART T-LIGHT HOLDER	4
5	23843	168469.60	1	United Kingdom	PAPER CRAFT , LITTLE BIRDIE	5
6	47566	139425.83	2568	United Kingdom	PARTY BUNTING	6
7	85099B	135834.66	3091	United Kingdom	JUMBO BAG RED RETROSPOT	7
8	84879	120182.05	2689	United Kingdom	ASSORTED COLOUR BIRD ORNAMENT	8
9	22086	114709.34	2011	United Kingdom	PAPER CHAIN KIT 50'S CHRISTMAS	9
10	23166	80575.63	228	United Kingdom	MEDIUM CERAMIC TOP STORAGE JAR	10
11	79321	79452.78	1133	United Kingdom	CHILLI LIGHTS	11
12	84347	68614.95	743	United Kingdom	ROTATING SILVER ANGELS T-LIGHT HLDR	12

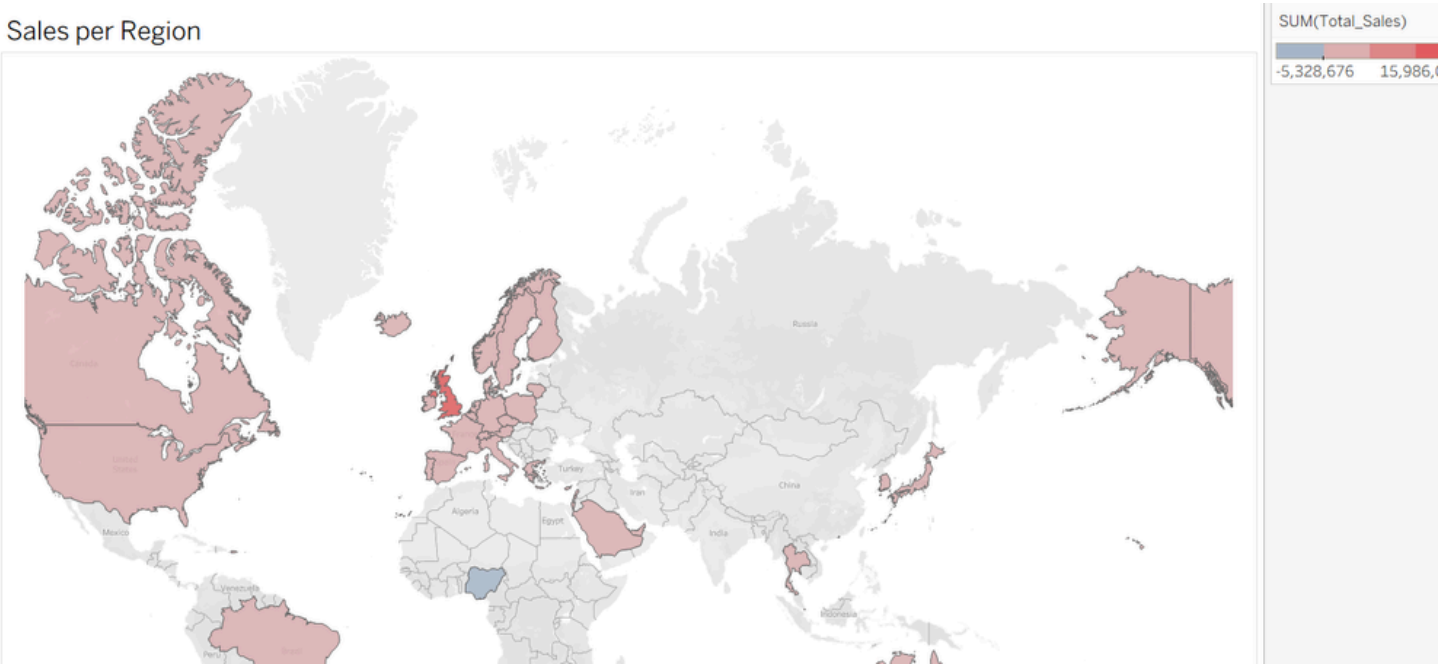
Challenges Encountered

There were several challenges encountered with the data

1. Negative quantities: This can affect several calculations and analysis. These cannot be spotted immediately as well.
2. Null Customer_ID: Having Nulls in Customer_ID made each row to not have a unique identity. These cannot be removed while cleaning as these show details of transactions

Suggest Improvements for Increasing Sales

Marketing can be sharpened, to attract customers from January to August. Discounts can be brought up in Nigeria to attract customers, or the sales have to pulled back dues to the sales falling in negatives. The European region, the higher focus can be given on the highest selling products



Identified Opportunities for Business Growth

The United States and the European States have a mid range sales so the sales strategies looked into for the United Kingdom can be put forward for the US and Europe. The sales can be introduced to countries adjacent to those which have mid-range sales. This would create new group of customers and hence increased sales.